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Quality: The Mayo Clinic Approach

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Developing highly reliable care for patients requires changes in some traditional beliefs of medical practice, an evolution toward a "system" of health care, the disciplined application of scientific principles, modifications in the way all future providers are trained, and a fundamental understanding by leadership that quality must become a business strategy and core work, not an expense or regulatory requirement. Quality at Mayo is defined as a composite of outcomes, safety, and service. A 4-part strategic construct focusing on Culture, Infrastructure, Engineering, and Execution has been developed to guide improvement activities and to ensure a comprehensive approach to better patient care. The Mayo Clinic experience has led to a greater understanding of the leadership commitment, organizational challenges, and the breadth of initiatives necessary to achieve highly reliable care. (Am J Med Qual 2009;24:428-440)

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> Give me but one firm spot on which to stand, and I will move the earth.
>
> —Archimedes[1]

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Work to improve quality (which we define as the composite of outcomes, safety, and service) at a health care institution can be a transformational lever . . . one that brings colleagues together on the common ground that attracted most of us to the profession. This common ground is *the firm spot on which we stand* together with Archimedes' lever.

We chose to write about our experience, not because we believe we have figured everything out (we have not) or because we are completely satisfied with our performance (we are not). We write about our experience because it has proved to be worthwhile and effective. We want to share what we have learned to date with others. Herein we will outline or detail principles, strategies, initiatives, change agents, and leadership actions. These have been identified as key components of effective organizational change, reliability, and safety.

Research, technological advancement, and new pharmaceuticals have resulted in longer survival over the past decades in conditions ranging from heart disease to cancer. Advances in the safety and reliability of health care *delivery*, however, have not kept pace. Health care in America has distressingly frequent errors and adverse events with harm to patients.[2] It delivers what we know to be the best care about as often as heads comes up when you flip a coin.[3] Patient safety and reliability of care delivery is an increasingly serious concern.

We have reason to believe that substantial improvements in reliability can be achieved in health care without the need for further research.[4] Several authors have identified that the science of quality improvement in health care may not be optimal, and there are efforts to better understand what makes quality improvement and *education* successful in health care.[5-7] We need only apply or implement what we already understand to be best practices to improve outcomes, safety, and service.[5-7]

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Herein, we review quality improvement methods that have been successful at Mayo Clinic.

THE MAYO CLINIC QUALITY CONSTRUCT

Our 4-part approach focuses on these 4 fundamentals: (*a*) optimizing our *Culture* for safety, outcomes, and service; (*b*) significantly enhancing supportive *Infrastructure*; (*c*) streamlining coherent *Engineering* efforts; and (*d*) delivering disciplined effective *Execution*.

Successful quality improvement at our institution is grounded in our mission to provide the best care to every patient every day through integration of clinical practice, education, and research. In this context, we organized our strategy and work under the guidance of this quality construct, which we have found to be of great value. It is adapted from one proposed by colleagues at the Institute for Healthcare Improvement.[8,9] Transforming medicine needs combined, measurable, disciplined efforts on all of these fronts (Figure 1).

CULTURE

There are many aspects of culture that must be addressed en face if improvement efforts are to be successful. Medicine cannot become highly reliable as long as autonomy, steep hierarchy, blame, independence, and opaqueness characterize an organization's culture.

Leadership must acknowledge that its organization must transform from a "cottage industry" to a standardized and interdependent delivery system. A basic tenet of a culture of safety is acceptance of a core of standard work based on best practice; this must be the rule and deviations from it are appropriate and expected only for patient-centered reasons, not ones that are physician-centered (eg, a warfarin regimen that is different from the organization's recommended standard because "this is how I like to practice"). A work environment with needless variation is one that increases the likelihood of medical errors by the health care personnel involved. There is an inherent tension between standardization to excellence and physician autonomy that must be understood and confronted.

A culture for safety is wholly compatible with innovation. Workflow must evolve from the heterogeneous cottage industry craftsperson era to a homogeneous platform of standard work. Innovation is then possible and expected such that change is directed toward quality improvement. If one does not have a common baseline, hypothesis-driven improvement is difficult because of a heterogeneous current state. We must engage the rigor of the scientific method to understand if we are truly making a difference in most improvement activities.

Many cultural changes can be made without expensive training programs and time away from caring for patients. Advances in behavioral competency, acceptance of standard work as the safest and most efficient practice, handoff/communication tools such as SBAR (situation-background-assessment-recommendations), and empowering process owners to spread best practices across the spectrum of care need not be expensive. They can be achieved by disciplined leaders with the fortitude to execute these important social system changes.

Integrated Group Practice

Colleagues at Mayo Clinic practice in a unique environment. We have an integrated multispecialty group practice with large education and research programs, all focused on improving the care of our patients. All Mayo Clinic physicians are salaried, which removes any personal financial incentive to perform a procedure or order an exam.[10-12] The patient is best served by the collective knowledge of the practice. Departments are accountable to the institution; all financial shortfalls or surpluses are shared with the integrated practice and redistributed by physician-led committees to achieve the institution's mission and vision.

An integrated medical practice with organized care coordination is an optimal ecosystem for learning. An integrated practice offers a community wherein the interests of medical staff, medical school, and hospital leadership are not set up as competing but are aligned. It is a structure in which inpatient–outpatient care is seen as a continuum. The hospital is considered not as a centerpiece but rather as a safety net when chronic and preventive outpatient care is unsuccessful.

Transparency

To educate our staff and employees about quality lapses and to energize them to find solutions, we have undertaken a formal policy of transparency and open communication.[13] It is a powerful, necessary, risky, and uncomfortable tactic. It is an antidote for complacency.

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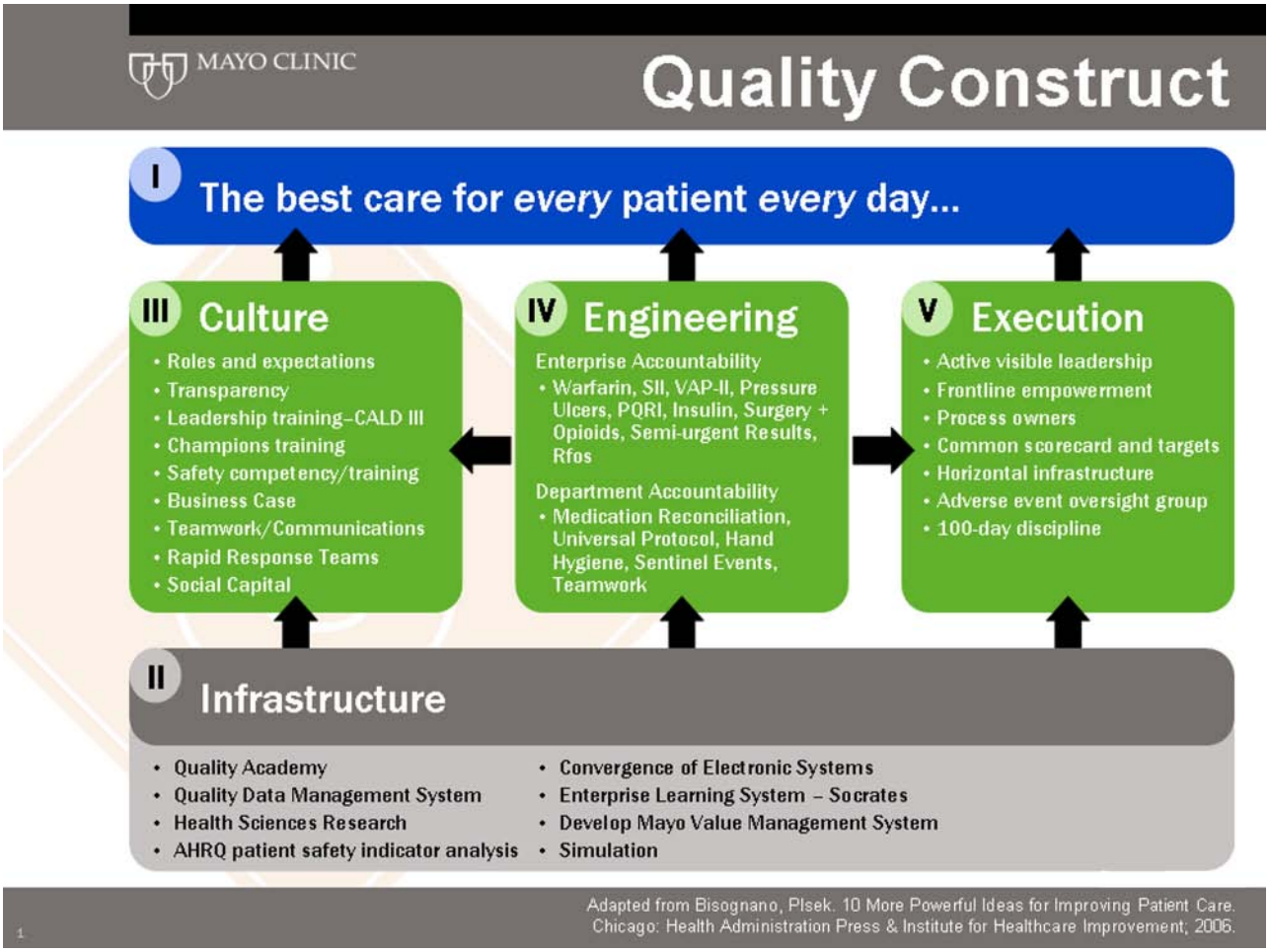

Figure 1. The Mayo Clinic Quality Construct. The 4 aspects of Infrastructure, Culture, Engineering, and Execution are aligned with our vision to provide the best care to every patient every day through integrated clinical practice, education, and research.[8,9]

The Mayo Clinic strategy and policy for advancing transparency include open disclosure with patients and deployment of internal and external Web sites with a broad range of quality metrics (http://www.mayoclinic.org/quality/). Beyond our Web presence, we send monthly e-mails to department leaders and post in outpatient and inpatient areas the latest outcomes, safety, and service performance of that unit relative to the rest of our practice. We also have regular features in newsletters and presentations with a tight linkage to our strategic plan. When a serious adverse event occurs in 1 of our 22 hospitals across 5 states, within a period of hours all relevant colleagues know of its occurrence. Transparency serves as the key to engage our top leaders and all members of the staff; it serves as a "wake-up call" when we fail to provide "the best care to every patient every day," accelerating and intensifying our efforts.

One of our group practices is located in Florida. In 2004, the Florida electorate overwhelmingly passed an amendment to the Florida Constitution: article X, section 25—the "Patient's Right to Know About Medical Incidents." Popularly known as Amendment 7, the legislation provides that any person who seeks to receive or has received health care treatment from a facility or provider has the right to request copies of documents that would otherwise have been privileged and confidential under other provisions of Florida law. This includes adverse incident reports and documents related to peer review, risk management, quality assurance, and credentials.

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Once a document is produced, there is no prohibition to the patient producing the document to the media. It is not clear, however, whether the document would be admissible in a medical malpractice trial. If the document is admissible, health care institutions may have greater difficulty defending a case.

Although Mayo Clinic in Florida has not experienced restrictions on its quality improvement activities as a result of Amendment 7, the courts have not ruled on whether participants in peer review and other proceedings may be deposed regarding their recollections of such meetings. The possibility of being deposed clearly could dampen the willingness of individuals to participate in such proceedings. However, our efforts to share the important lessons learned across all 5 states wherein Mayo Clinic cares for patients is supported and encouraged by leadership.

Rapid Response System

Although rapid response systems are part of our infrastructure, we include them in the Culture section because, more than anything else, our intent with them is to change the way in which we interact with each other on behalf of our patients.

Nationwide, patients who suffer cardiopulmonary arrest in hospitals frequently (~70%) have identifiable signs of impending collapse for hours prior to arrest.[14-16] In such cases, the attending physician is notified only ~25% of the time.[15] A rapid response system is primarily intended to address care shortfalls like these that arise from failure to recognize or failure to rescue.

A rapid response system has as its centerpiece a team of clinicians who bring critical care expertise to the bedside, with the goal of averting a cardiorespiratory code team call.[17] The primary intent is to reduce preventable mortality from failure to recognize or rescue. Two important secondary goals are the education of staff in acute care medicine and, most important, the generation of a cultural change that increases psychological safety. An environment of psychological safety is one in which asking for help is not considered a sign of weakness; all members of the patient care team must be comfortable speaking up when they have a concern. It is one wherein "hierarchy and turf" do not impede a quick response and there is no fear of verbal abuse for sincerely acting on behalf of the patient, even if one's judgment eventually is shown to be wrong.

Our cross-functional Rapid Response Teams are trained together in our Simulation Center. Their curriculum is updated regularly with findings from our mortality review (deaths at Mayo Clinic are reviewed by a multidisciplinary team of physicians and nurses and their findings are shared with leadership monthly).

"Plus One" is an advanced version of a rapid response system. It is a carefully designed communication program that provides tools to facilitate effective communication in a variety of situations in addition to emergencies. Essentially, the Plus One program empowers all health care providers to call for help. It creates psychological safety. Plus One outlines the best way to clearly convey information to get needed assistance and demonstrates that it is not only acceptable but encouraged to pursue issues up the chain of command if one does not receive the immediate action needed. In the first chapter of *Nursing Against the Odds*, Suzanne Gordon[18] illustrates the reality of the health care structure. New, inexperienced providers (physicians) are frequently the first responders for patient care needs. However, differences in knowledge and assessment skills can be great between nurse and physician. When simple resolution of these differences does not occur, the Plus One program empowers the nurse to add more experienced participation so that the conflict does not linger—at the patient's expense. The programmatic aspect of Plus One includes review of Plus One documentation by institutional leadership. This allows review, analysis, and trending of communication difficulties; permits ongoing education; helps team building; and goes beyond "near emergencies"—advantages not present in a basic rapid response system.

Culture of Safety

Culture of Safety includes attributes of teamwork, trust, mutual respect, communication, emotional intelligence, transparency, and just culture. In a just culture, there is acknowledgment of the propensity of good, competent, conscientious colleagues to human error. A sense of justice must work in concert with the demand for accountability. Frontline workers must feel comfortable disclosing errors, including their own. At the same time, we must maintain professional accountability. Health care culture has traditionally held individuals accountable for all errors or events that affect patients under their care. A just culture

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recognizes that individual caregivers should not be held accountable for system failures over which they have no control.[19] A just culture recognizes that many errors represent predictable consequences of human beings working in imperfect systems. However, risky and reckless behavior must be addressed with coaching and punishment, respectively.[20-22]

Social Capital

Social capital is the active connections among colleagues caring for patients. To reach high reliability, health care must foster systems change to drive continuous learning. A key component of a comprehensive strategy involves the conscious investment in social capital. Social capital investments move an organization from a collection of individuals toward an agile, coherent collective mind, a form of intelligence that emerges from the collaboration of many individuals.

For high reliability, there must be nimble and effective diffusion of best practices and safety nets. Optimal organizational learning requires fluent (*a*) intrateam; (*b*) interteam, intrasite; and (*c*) interteam, intersite communication. The networks must be purposely engineered and nurtured (eg, identify a quality leader in every department at each site and connect them via meetings, communication, information technology). A fundamentally important tactic in this cultural transformation is the training of health care's most inexperienced learners—medical and nursing students, residents—*together* on cross-functional teams.[23] Building and sustaining social capital is integral to high reliability.

A Team Approach

The medical profession must embrace the fundamentals of teamwork to make substantive improvements in reliability. Teamwork directly relates to errors of communication and handoffs, our most common root causes of sentinel events.

Delivery of health care to an individual patient requires carefully integrated services of a large number of individuals, including not only physicians from diverse disciplines, nursing, pharmacy, and laboratory staff, but also receptionists, secretaries, and educators, among others. Some groups, such as nursing, have a rich history of working closely together as a team. Although physicians are trained primarily to function autonomously, optimal care dictates that they work interdependently on cross-functional teams. In viewing culture change, Schein et al[14] discuss the events that shape or change culture—situations that are accompanied by success where the social group, body, organization, or team adopt or change existing practice because of the success. As part of our effort toward changing/improving the culture to continually focus on safety, we have dedicated resources to emphasize internal transparency. Our teams' successes are publicized in many formats (eg, intranet, alerts, announcements). The emphasis on sharing this information helps ensure that everyone feels part of the team.

One of our tactics is to measure teamwork, trust, mutual respect, and communication, particularly from the perspective of nonphysician members of our cross-functional work units. The most common cause of sentinel events, according to the Joint Commission, is communication breakdowns.[24] Work units with optimal teamwork and communication scores are safer patient care areas. We must measure and manage what is meaningful to us and what we want to improve. We have added a work unit teamwork metric to our performance scorecards. A high reliability culture makes it easy to learn from its mistakes. If colleagues have trust, feelings of teamwork, mutual respect, and communication on their team, they can learn from others' mistakes and improve the outcomes and safety of patients.

Effective communication is the cornerstone for building high-reliability teams with a goal to deliver safe care and superior outcomes. A Harris poll conducted for *The Wall Street Journal* found that patients judged their physicians' interpersonal skills and listening as more important than good medical judgment. Building trustworthy relationships with patients as they enter a medical institution for the first time pays dividends throughout the patient's health care journey. Impeccable communication between staff leads to improved employee satisfaction and stronger teams, resulting in safer care, enhanced outcomes, and employees who have loyalty to each other and the organization. A learner-centered and skills-based program was developed at Mayo Clinic with an initial targeted audience of physicians. Physician faculty received training from the American Academy for Communication in Healthcare, with goals to develop a 4-hour curriculum that would be acceptable to all physicians, regardless of specialty,

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and to learn facilitation skills to deliver the skills-based course to their physician colleagues. Using the goal of "excellence" for quality of care, the proportion of excellent ratings for overall quality of care increased from 74.0% in 2005 (the start of the communication curriculum) to 76.4% in 2007 (after all physicians participated in the curriculum; $P < .001$) at Mayo Clinic Florida. The proportion of excellent ratings for the physicians' quality of care increased from 71.3% in 2005 to 74.0% in 2007 ($P = .004$). Patients who would definitely recommend the institution to family and friends increased from 88.2% in 2005 to 90.2% in 2007.

To improve the safety and reliability of clinical handoffs, we have adopted the SBAR technique and have found that it provides an effective framework for communication between members of the health care team about a patient's condition. SBAR is a simple, concrete tool that is useful for framing conversations, especially critical ones, requiring a provider's immediate attention and action. SBAR sets expectations for what will be communicated. It defines how best to communicate between members of the team, which is essential for developing teamwork and fostering a culture of patient safety.[25]

As part of Team Approach, we have focused on teamwork as a universal initiative at all sites and have fostered "Team Training" through our Quality Academy. The benefit of teams in quality improvement has been validated by others.[26]

Safety Competency

We start with the premise that it is reasonable to expect that every health care provider who performs a procedure should have documented safety competence. This is our goal, not the status quo. The fact that a physician is board certified in his or her specialty is insufficient; confirmation of competence in a simulation center, followed by observation and sign-off are more reassuring from the patient's perspective. For example, Mayo Clinic Rochester residents and fellows must first demonstrate competency in central line placement in our Simulation Center before performing this procedure on a patient. We have a "Surgical Olympics" in the Simulation Center to demonstrate surgical competency.

Any new resident or fellow must successfully complete a Web-based safety module that introduces the basics of patient safety (eg, universal protocol, SBAR, hand hygiene, sentinel events, appropriate reporting) *before* being permitted to care directly for patients. This is the first mandatory orientation message that these caregivers receive, an important statement in and of itself. All medical students complete a quality improvement curriculum and many participate in an elective wherein they are part of a process improvement team. All internal medicine residents must successfully complete an approved improvement project to graduate. This in an investment in our institution's future, as Mayo Clinic hires a large percentage of graduating residents for staff positions.

Demonstrated competency in a simulated environment is setting a new bar. No longer should we accept level of training or seniority as sufficient to perform a procedure that puts a patient at risk. We must be prepared to limit the number of practitioners so that defect rates, infection rates, and complication rates, among others, reach defined standards of excellence. This certification must also include an understanding and perhaps a measure of the appropriateness of the procedure. As we continue to add newer invasive interventions that carry significant risks and costs, we are obligated to understand when they are beneficial and when they are evidence based. Leadership requires making difficult and at times controversial decisions that may infringe on the autonomy of health care providers.

Business Case

Part of building the will for change is making a business case for improving outcomes, safety, and service, even in a not-for-profit enterprise. We work to improve quality because it saves lives or decreases hospital-acquired pneumonias. But when we do it right, it saves money. The process/system becomes leaner and needless waste, variation, and defects are driven out. It intentionally evolves into a system that is safer, more efficient, and more profitable. Productivity gains come from dividends inherent in reducing rework due to poor quality, reductions in malpractice costs, revenues from pay-for-performance programs, and reductions in cost structure that drive waste and variation from practice.

Reducing waste and variation is the primary source of cost savings that leads to improving the value of our care (Value = Quality [Outcomes, Safety, Service]/Cost). Defective care causes considerable harm and is costly. Suboptimal delivery of appropriate preventive services, evidence-based, acute, and chronic care is also defective and globally

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costly. The goal is to make the case for quality as a business strategy, not an expense.[27]

Organizations need to appreciate that quality is no longer a mandate from external agencies or simply a program of continuous improvement. It needs to become a core business strategy for any health care organization's ability to survive and thrive.

It is important to measure the financial impact of improvement work. Colleagues from our department of finance assessed projects conducted by 14 quality academy teams and determined that the internal rate of return was 85%. Most of these projects were focused on quality, not efficiency.

INFRASTRUCTURE

Addressing Culture is necessary but insufficient. We also must have supporting infrastructure to raise our reliability beyond the limits of human performance.

Quality Academy

Historically, quality improvement has rarely been part of the medical school or other professional health care curriculum. Colleagues whose primary responsibility is quality improvement (with expertise in a variety of improvement modalities) have not, until recently, been a fixture at most academic medical centers. We seek to have improvement in the processes of care (and attendant safety nets) be part of one's core work of caring for patients, not a part-time, spare-time endeavor. Furthermore, leadership must be conversant with the science and potential of such methods. For these reasons, we established the Quality Academy.

The Quality Academy comprises several key programs, including the following:

1. Two-day *Leadership Program* designed for top institutional leaders.
2. Eight-hour *Champions Course* designed to teach leaders how to initiate improvement efforts and provide the appropriate support.
3. Nine-day *Teams Training Course* spread over 100 days that focuses on teaching our Value Management Systems' DMAIC (ie, Define, Measure, Analyze, Improve, and Control) tools and completing a project. This module is modeled after the Advanced Training Program at Intermountain Health led by Dr Brent James.[28]
4. *Rapid Process Improvement Workshop* (RPIW), a week-long improvement team effort that is a shorter version of the Teams Training Course.
5. *Annual Conference on Quality* with more than 1000 Mayo Clinic attendees (in person and virtual).
6. *Annual Human Factors Postgraduate Course.*
7. Monthly *Quality Forum* with internal and external speakers.

Electronic Records

In the early 1990s, Mayo Clinic made a considerable commitment to have electronic medical records at all sites. At the same time we also began the process of converting all medical images (eg, X-rays, electrocardiograms [ECGs]) to electronic formats. Currently, patient information is electronically available to practitioners at all sites across the outpatient and inpatient continuum. Although the efficiency of these various systems has great room for improvement, our patients now benefit in immeasurable ways as a result of these investments. Mayo is now leveraging this environment by implementing numerous rule sets that improve patient safety through electronic verification of items such as patient identification, drug/allergy interactions, and blood transfusions.

Point-of-Care Information

Medical errors harm patients. We can also harm patients because of a knowledge gap. We are using information technology to push knowledge to caregivers who do not know what they do not know, so that "Mayo knows what Mayo knows."

Closed-loop communication of urgent and semi-urgent results is a hallmark of a highly reliable practice. For instance, when an outpatient has a long QT interval identified on an ECG, the provider who ordered the exam is notified electronically and an acknowledgment is monitored. The notification includes a hyperlink to information regarding our long QT outpatient clinic as well as the optimal care, dangerous drug interactions, and other pertinent data. Pushing this information to providers at the point of care promotes state-of-the-art care and standards.

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Simulation Center

Another major component of our strategy includes Mayo Clinic's multidisciplinary 10 000 ft^2 Simulation Center. Our program optimizes a new paradigm in medical education by offering "experiential learning" using 3 modes of simulation:

- Standardized patients—actors who play the role of patients
- Task Trainers—virtual reality "video games"
- High-tech mannequins—surrogate patients of a robotic nature

In the Simulation Center, health care providers learn and become certified in procedures and activities that, if performed while learning in the "production environment," would put patients at unnecessary risk for adverse events with harm. Competencies include airway fire management, teamwork simulations, central line placement, and codes, among others. More than 6000 learner encounters occurred in the Simulation Center in 2007.

Management of Quality Information

The response to the quality chasm has been a rising tide of national concern regarding quality in American medicine, accompanied by an ever-increasing number of requirements, standards, and reporting. The need to meet multiple complicated process and reporting measures requires considerable time and effort. Uncertainty remains as to which of the many measures truly reflects outcomes, safety, and reliability. We, along with other health care organizations across the country, are confronted by the dilemma that time and energy required for mandated external reporting easily diverts attention from the ongoing systems and culture work that is so fundamental for achievement of true and lasting change. Efficient and effective management of quality information is a critical success factor for health care in America.

Almost every health care institution has a set of comprehensive financial applications that allow them to bill, report, and analyze their financial status. However, few if any have comprehensive quality systems with a fully developed set of outcomes, safety, and service measures that allow them to understand the status and measure the effect of changes in quality activities across their organizations. Developing this type of system is another core element of Mayo Clinic strategy. Information infrastructure is a critical aspect of understanding and catalyzing change. Our plan includes a common events database and common project database for all 15 medical centers of Mayo Clinic.

Health Sciences Research

Partnering with research colleagues is a critical cornerstone for our transformational goals. Our Health Sciences Research colleagues are partners who help us understand the highest impact opportunities for reducing mortality and adverse events with harm. Rigorous observational and statistical analysis is necessary to validate or refute quality improvement initiatives. In leveraging research as a partner, we emphasize the publication potential for individual academic career advancement. The science of quality and health care delivery is a valid academic career track. There are many venues for peer-review publication of quality-related work.[29,30]

Health Sciences Research partnership allows us to understand the patient care and economic opportunities. For example, the approximate median incremental cost for Mayo Clinic adverse events with harm is $5060 (mean incremental cost = $18 346; J. M. Naessens, C. R. Campbell, B. P. Berg, et al, in press, June 2009). The median incremental length of stay for adverse events with harm at Mayo Clinic is approximately 2.1 days (mean incremental length of stay = 6.8 days). These data are from an analysis of 5883 adverse events with harm (vast majority of events were minor). Approximately 4% of discharges from Mayo Clinic hospitals have an adverse event with harm. By projection, a 10% reduction in the event rate would result in $6.4 million of net cost savings. Information like this allows us to make both the patient care and business case with our own data and to focus our improvement efforts on high-impact areas.

ENGINEERING

Addressing Culture and Infrastructure are necessary but insufficient. We must also promote improvement efforts using the best that the disciplines of engineering and human factors have to offer. We look at engineering as the disciplined application of scientific principles to creatively develop or redesign effective, efficient, safe, and reliable systems to enhance the care of our patients.

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Effective Use of Systems and Human Factors Engineering

Unlike other industries, such as aviation, most areas of medicine have not applied systems or human factors engineering to the delivery of health care in a systematic and disciplined way.[31] We see needless waste, variation, and defects in our care not because of our people but because of our systems. We need systems in place that protect patients from the scientifically understood human factors that result from hard-working, conscientious, competent health care providers falling short of their aspirations every day. Mayo has had a group of engineers, analysts, and improvement experts called Systems and Procedures since 1942. They helped establish many standardized practices such as the development of a single consolidated medical record system across the inpatient and outpatient environments, common appointment systems, and desk systems.

We must design high-reliability systems. The best aircraft carriers and some nuclear power plants are considered high reliability.[32,33] Within health care, this same type of reliability has been achieved within some portions of anesthesia and blood banks.

Caring for patients is not the same as making widgets or flying jets. But what is inherently more complex about administering a medication or labeling a radiograph or specimen than landing an F-16 on a moving, wet runway in the middle of the South Pacific? One can break down medical care into microprocesses that together comprise a system. Each step can be analyzed and maximized for reliability and efficiency.

Value Management System

As our quality strategy was refined in 2006, we found there were many quality improvement efforts under way across Mayo Clinic sites. Yet we also learned that there were numerous approaches and that diffusion of results was limited. Most efforts were not fully resourced and there was suboptimal communication between teams. Some improvement efforts had been going on for years without the necessary resources or discipline to come to closure.

Critical for success is defining a specific improvement strategy. We researched, developed, and refined the Mayo Clinic Value Management System, which is characterized by focus, discipline, and the effective use of data. It uses a DMAIC umbrella, a well-known series of standardized steps, that is supported by a diverse set of evidence-based engineering and human factors tools and principles from Six Sigma, LEAN, Theory of Constraints, Deming, Rapid-Cycle Plan-Do-Study-Act, Juran, and Queuing Theory, among others. The intent has been to define an improvement model that fits with our culture and that will be sustainable over time. It is important that colleagues at all sites speak the same language of improvement.

Each improvement team has a project charter that is typically organized in "100-day" segments to ensure that measurable targets are in place with timely expectations. Each team is led by a process owner (usually a health care provider) and a project manager. Additionally, the teams are staffed with process experts who have skills in our value management system, measurement coordinators, content experts, and nursing and line administrators to assist with implementation.

We believe this approach to be a time-tested and industry-validated approach to care improvement for our patients. It looks at the cultural readiness, a serious way of moving forward with discipline and accountability, and it starts with a plan and a commitment to fully staff efforts. The discipline of focus on an important problem, measuring improvements, fully implementing proven results, closing the project, and moving on to our next priority has been effective.

Our current practice is to assemble a limited number of fully staffed teams that focus on the most important, high-priority, high-impact targets. These initiatives are vetted and developed by quality and practice leadership committees. Our initial teams in 2006 were organized around what we called the "compulsory elements" from outside agencies such as the Centers for Medicare and Medicaid Services (CMS). We were able to demonstrate substantive improvement (ie, moving from midlevel to one of the leading academic medical centers) in our compliance with process measures such as acute myocardial infarction, surgical infection prevention, pneumonia, and heart failure. This led us to direct our attention to other potentially higher impact opportunities for lowering mortality and adverse events with harm, as well as to optimize delivery of preventive services and evidence-based acute and chronic care processes. Other opportunities include our top 5 causes of failure to rescue, our top 5 causes of needless death from our mortality reviews, and our top 5 causes of malpractice. We track and manage teams at 3 primary levels.

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1. *Enterprise teams.* We have worked across the enterprise (in 5 states and at 15 medical centers) on systemwide initiatives such as pneumonia, critical and semiurgent result reporting, retained foreign surgical objects, pressure ulcers, heart failure, and warfarin management.
2. *Site teams.* In addition to priority initiatives targeted across all Mayo Clinic sites, each site has its own local priorities that are organized and managed with the same value management system. Results are expected to be shared with appropriate staff at other sites.
3. *Departments.* We expect all 29 departments and their major divisions to have quality metrics and conversations across campuses. Most activities are site-specific, but in 2008, we asked all departments to work together on a core set of 5 safety metrics (ie, universal protocol, hand hygiene, medication reconciliation, teamwork, and sentinel event management). Departmental efforts are many times the impetus for a project being chartered to attend the Quality Academy Teams Training.

Today we have more than 100 system engineers and improvement experts at Mayo Clinic. Most reside in a central work unit and deploy based on institutional priorities. Our long-term goal is to have most of this expertise work and live in departments and divisions—to have this be part of our institutional "fabric." Some departments hire black belts and system engineers and support them with project managers and data experts.

EXECUTION

Addressing Culture, Infrastructure, and Engineering are necessary but insufficient. We must also execute.

Mayo Clinic quality operations are led by an enterprise quality committee that is composed of leadership from each of the 4 entities (Mayo Clinic Rochester, Mayo Clinic Florida, Mayo Clinic Arizona, and the Mayo Health System). Each entity has a Quality Oversight Committee that is composed of leaders who have responsibility for areas such as safety, service, outcomes, measurement, accreditation, and transparency. These committees serve as the surrogate leader for quality work throughout the institution. They are accountable to the practice committees and boards. Each group is responsible for working with departments and hospital staff to execute priorities.

Quality Dashboard Setting Performance Expectations

We have established a systemwide electronic dashboard with common data definitions and targets for each measure. The use of standardized reporting across entities with targets, external benchmarks, and drilldown capabilities serves as a basis to track progress and build will. Transparency is not effective without trusted, comparable, reliable data. We must insist on *P* values and control charts as we analyze data and pursue improvement.

Transparency and a common dashboard for all of Mayo Clinic are fundamental components of execution. Visibility and goal-oriented management of meaningful system-level outcomes, safety, and service metrics is central to improvement, spread, and maintenance of gains. Our dashboard includes metrics for each site that allow comparison and electronic interrogation for details behind high-level values. Drilldown capability allows for analysis at campus, department, division, and work unit levels. The dashboard comprises 3 measurement categories:

1. Outcomes
   a. Hospital standardized mortality ratio
   b. Agency for Healthcare Research and Quality (AHRQ)—death in low mortality diagnosis related groups (DRGs)
   c. Percentage initiatives in green status
   d. Core measure process composite (all or none)
   e. Adjusted inpatient mortality—acute myocardial infarction
   f. Adjusted inpatient mortality—heart failure
   g. Adjusted inpatient mortality—pneumonia
   h. Catheter-related bloodstream infections
2. Safety
   a. Reportable adverse health care events
   b. Days away, restricted time, and job transfer episodes per 100 full-time employees
   c. Patient events with harm

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   d. Medication events with harm/1000 inpatient days
   e. AHRQ-Patient Safety Indicator Composite
3. Service
   a. Patient satisfaction—percentage excellent in overall quality of care rank

Data are reported monthly for all of Mayo Clinic and also by site. Control charts are used whenever possible. The dashboard is color coded to provide at-a-glance assessment of progress toward goals. Performance information is available to all Mayo Clinic employees online (Figures 2-6).

Mayo Clinic consistently scores at or near the top of major quality rating groups including the Dartmouth Atlas, *US News and World Report*, University HealthSystem Consortium, The Leapfrog Group, Health Grades, and CMS Quality Rankings. Recently, the Network for Regional Healthcare Improvement published a ranking of the 73 largest health systems in America using a combination of HCAHPS (Hospital Consumer Assessment of Healthcare Providers) data and CMS core measure scores. Mayo Clinic rated second among the 73 (www.NRHI.org/news.html).

Diffusion

For optimal technology management and high-reliability patient care, there must be nimble and effective diffusion of best practices and safety nets. This must occur both within an organization and throughout the nation. Wide diffusion of best practices for technology use has proved slow and inconsistent.[34-36] The identification of a process owner and project champion has facilitated diffusion.[37]

CONCLUSION

To deliver high-reliability health care, organizations must create and nurture a collective mind. The challenge requires optimal teamwork and communication skills, an attitude of transparency, cross-functional work, and improvement teams with psychological safety and technology that pushes the best and safest practice information to those at the point of care.

High-reliability health care requires 4 transformative vectors: Culture, Infrastructure, Engineer-ing and Execution. All are necessary; none stands alone.

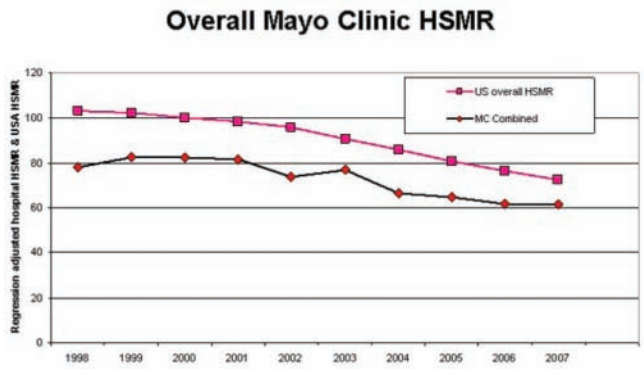

Figure 2. Mortality results. Mayo Clinic hospital standardized mortality rate (HSMR) benchmarked against overall US performance.

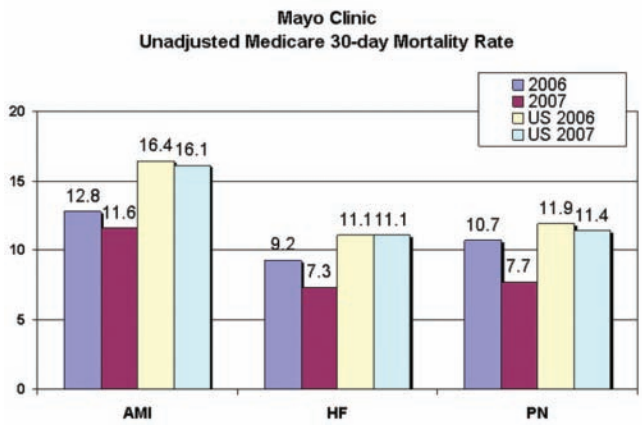

Figure 3. Thirty-day mortality results for acute myocardial infarction (AMI), heart failure (HF), and pneumonia (PN), benchmarked against US performance in 2006 and 2007.

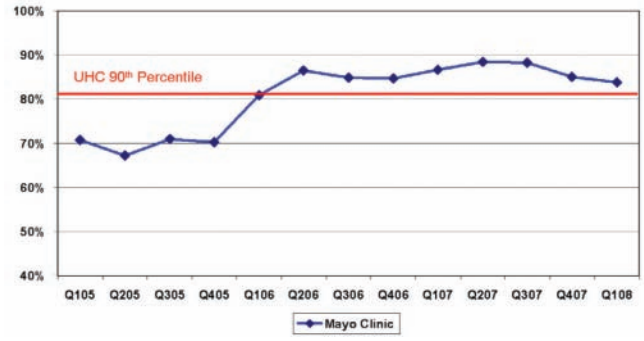

Figure 4. Summary of Centers for Medicare & Medicaid Services core measures. Data are displayed as all or none. Note improvement over 3-year period across all Mayo Clinic sites.

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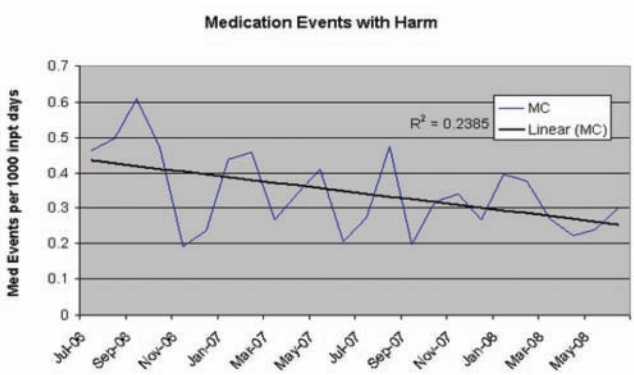

Figure 5. Medication events with harm. Note statistically significant reduction in medication events with harm across all Mayo Clinic sites over a 2-year period.

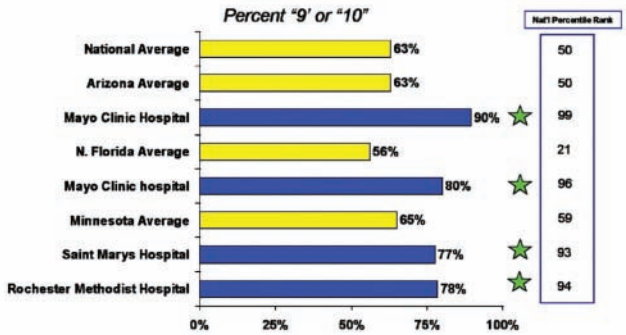

Figure 6. Centers for Medicare & Medicaid Services HCAHPS patient satisfaction data April-June 2008. Mayo Clinic hospitals rank in the 93rd to 99th percentiles. HCAHPS, Hospital Consumer Assessment of Healthcare Providers.

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REFERENCES

1. Archimedes. Quoted by Pappus of Alexandria. *Collection* (*Synagoge*, Book VII, c. AD340). Berlin, Germany: Hultsch; 1878:1060.
2. Kohn LT, Corrigan JM, Donaldson MS. *To Err Is Human: Building a Safer Health System*. Washington, DC: National Academies Press; 1999.
3. McGlynn EA, Asch SM, Adams J, et al. The quality of health care delivered to adults in the United States. *N Engl J Med*. 2003;348:2635-2645.
4. Amalberti R, Auroy Y, Berwick D, Barach P. Five system barriers to achieving ultrasafe health care. *Ann Intern Med*. 2005;142:756-764.
5. Robert Wood Johnson Foundation. Evidence for improvement: evaluating quality improvement training programs. http://www.rwjf.org/applications/solicited/cfp.jsp?ID=20282. Accessed May 29, 2009.
6. Boonyasai RT, Windish DM, Chakraborti C, Feldman LS, Rubin HS, Bass EB. Effectiveness of teaching quality improvement to clinicians. *JAMA*. 2007;298:1023-1037.
7. Peterson ED. Optimizing the science of quality improvement. *JAMA*. 2005;294:369-371.
8. Provost L, Miller D, Reinertsen J. *A Framework for Leadership of Improvement*. Cambridge, MA: Institute for Healthcare Improvement; 2006. http://www.ihi.org/IHI/topics/leadingsystemimprovement/leadership/emergingcontent/aframeworkforleadershipofimprovement.htm. Accessed May 29, 2009.
9. Bisognano M, Plsek P. *10 More Powerful Ideas for Improving Patient Care*. Chicago, IL: Health Administration Press; 2005.
10. Hillman BJ, Joseph CA, Mabry MR, Sunshine JH, Kennedy SD, Noether M. The frequency and costs of diagnostic imaging in office practice: a comparison of self-referring and radiologist-referring physicians. *N Engl J Med*. 1990;323:1604-1608.
11. Hillman BJ, Olson GT, Colbert RW, Bernhardt LB. Responses to a payment policy denying professional charges for diagnostic imaging by nonradiologist physicians. *JAMA*. 1995;274:885-887.
12. Hillman BJ, Olson GT, Griffith PE, et al. Physicians' utilization and charges for outpatient diagnostic imaging in a Medicare population. *JAMA*. 1992;268:2050-2054.
13. Swensen SJ, Cortese DA. Transparency and the "end result idea." *Chest*. 2008;133:233-235.
14. Schein RMH, Hazday N, Pena M, Ruben BH, Sprung CL. Clinical antecedents to in-hospital cardiopulmonary arrest. *Chest*. 1990;98:1388-1392.
15. Franklin C, Mathew J. Developing strategies to prevent in-hospital cardiac arrest: analyzing responses of physicians and nurses in the hours before the event. *Crit Care Med*. 1994;22:224-247.
16. Buist MD, Moore GE, Bernard SA, Waxman BP, Anderson JN, Nguyen TV. Effects of a medical emergency team on reduction of incidence of and mortality from unexpected cardiac arrests in hospital: preliminary study. *BMJ*. 2002;324:387-390.
17. Bellomo R, Goldsmith D, Uchino S, et al. A prospective before-and-after trial of a medical emergency team. *Med J Aust*. 2003;179:283-287.
18. Gordon S. *Nursing Against the Odds: How Health Care Cost Cutting, Media Stereotypes, and Medical Hubris Undermine Nurses and Patient Care*. Ithaca, NY: Cornell University Press; 2005.
19. Wiegmann DA, ElBardissi AW, Dearani JA, Daly RC, Sundt TMI. Disruptions in surgical flow and their relationship to surgical errors: an exploratory investigation. *Surgery*. 2007;142:658-665.
20. Reason J. Human error: models and management. *BMJ*. 2000;320:768-770.
21. Marx D. *Patient Safety and the "Just Culture": A Primer for Health Care Executives*. New York, NY: Columbia University; 2001.
22. Rozich JD, Howard RJ, Justeson JM, Macken PD, Lindsay ME, Resar RK. Standardization as a mechanism to improve safety in health care. *Jt Comm J Qual Patient Saf*. 1994;30:5-14.

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23. Leonard M, Graham S, Bonacum D. The human factor: the critical importance of effective teamwork and communication in providing safe care. *Qual Saf Health Care*. 2004;13:85-90.
24. The Joint Commission. Sentinel event. http://www.jointcommission.org/sentinelevents/. Accessed May 29, 2009.
25. Institute for Healthcare Improvement. SBAR technique for communication: a situational briefing model. http://www.ihi.org/IHI/topics/patientsafety/safetygeneral/tools/SBARTechniqueforCommunicationASituationalBriefingModel.htm. Accessed May 29, 2009.
26. Marsteller JA, Shortell SM, Lin M, et al. How do teams in quality improvement collaboratives interact? *Jt Comm J Qual Patient Saf*. 2007;33:267-276.
27. Lee LH, Swensen SJ, Gorman CA, Moore RR, Wood DL. Optimizing weekend availability for sophisticated tests and procedures in a large hospital. *Am J Manag Care*. 2005;11:553-558.
28. Intermountain Healthcare. Institute for healthcare delivery research. http://intermountainhealthcare.org/about/quality/institute/Pages/home.aspx. Accessed May 29, 2009.
29. Finkielman JD, Morales IJ, Peters SG, et al. Mortality rate and length of stay of patients admitted to the intensive care unit in July. *Crit Care Med*. 2004;32:1161-1165.
30. Gajic O, Afessa B, Hanson AC, et al. Effect of 24-hour mandatory versus on-demand critical care specialist presence on quality of care and family and provider satisfaction in the intensive care unit of a teaching hospital. *Crit Care Med*. 2008;36:36-44.
31. Shortell SM, Singer SJ. Improving patient safety by taking systems seriously. *JAMA*. 2008;299:445-447.
32. Weick KE, Sutcliffe KM. *Managing the Unexpected: Assuring High Performance in an Age of Complexity*. New York, NY: John Wiley; 2001.
33. Keroack MA, Youngberg BJ, Cerese JL, Krsek C, Prellwitz LW, Trevelyan EW. Organizational factors associated with high performance in quality and safety in academic medical centers. *Acad Med*. 2007;82:1178-1186.
34. Ting HH, Shojania KG, Montori VM, Bradley EH. Quality improvement: science and action. *Circulation*. 2009;119:1962-1974.
35. Wennberg JE, Fisher ES, Skinner JS, Bronner KK. Extending the P4P agenda, part 2: how Medicare can reduce waste and improve the care of the chronically ill. *Health Aff (Millwood)*. 2007;26:1575-1585.
36. Wennberg JE, O'Connor AM, Collins ED, Weinstein JN. Extending the P4P agenda, part 1: how Medicare can improve patient decision making and reduce unnecessary care. *Health Aff (Millwood)*. 2007;26:1564-1574.
37. Dedhia P, Kravet S, Hess R, et al. Facilitators and barriers involved in conducting hospital-based research across multiple sites. *J Clin Outcomes Manag*. 2008;15:235-239.

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